Asu Singh

+91-9910521365 | itsasusingh@gmail.com | linkedin.com/in/asusingh | Delhi, India

PROFESSIONAL SUMMARY

Software Development Engineer with 5+ years building production financial-services software, approaching six years, and hands-on experience using and building agentic AI systems for personal projects. Strong foundation in Java, Python, event-driven systems, performance engineering, database reliability, observability, and machine learning. Published peer-reviewed research in Cognitive Computation on a Hybrid Social Group Optimization method for COVID-19 chest X-ray detection, achieving over 99% accuracy. Interested in applying production engineering, empirical evaluation, and practical AI-system building to reliable, beneficial AI deployments.

TECHNICAL SKILLS

Java & Backend APIs: Java 8+, Spring Boot, Microservices, REST APIs, Hibernate
Systems & Reliability: Multithreading, Performance Tuning, Event-Driven Systems, MQ-like Messaging, Production Debugging, Log Analysis
Data & Observability: Oracle, PostgreSQL, Relational Databases, AWS, Grafana, Prometheus, Plotly
Engineering Practices: Test-Driven Development, JUnit, Mockito, Agile, Unix
Web: Django, HTML5, CSS, JavaScript
AI & ML: Agentic AI, Machine Learning, Computer Vision, Empirical Evaluation

WORK EXPERIENCE

ION Trading

Aug 2020 - Present

Senior Software Development Engineer

Noida, India

- Designed and developed XTP, ION's real-time back-office trading product, used in production by major trading firms and banks.
- Led a Java multithreading refactor that increased trade-processing performance by 20x during end-of-day processing.
- Developed Alazium, a log-analysis initiative that parses logs exceeding 100 GB and visualises workflows with Plotly, reducing production issue-analysis time by 30%.
- Designed a recovery feature for halted end-of-day processing, enabling automatic recovery from fail-fast scenarios and reducing downtime by 90%.
- Worked with XTP's in-house MQ-like, Kafka-like event-driven notification system, understanding its internal workings and operational failure points.
- Redesigned historical-data cleanup using a tailored linear-regression approach, improving database performance and reducing processing time by 95% under load.
- Led migration of XTP CI systems from Red Hat 7 to Red Hat 8, supporting reliable engineering delivery.

Holisol Logistics

Jun 2019 - Sep 2019

Software Development Intern

Delhi, India

- Developed a Daily Run Sheet Generator with a Django frontend to optimise deliveries and presented the application to the company founder for deployment approval.

SELECTED PROJECTS & PUBLICATION

Agentic Personal Assistant — Applied AI

Built a personal assistant using free NVIDIA models and a coding harness. It maintains ideas and notes through natural-language chat, helps plan tasks and days, and is accessible from a phone through messenger apps 24/7.

Free NVIDIA Models, Coding Harness, Messenger Apps

Wardrobe AI — Personal Project

Built a personal wardrobe assistant that lets users scan existing photos to automatically create a digital wardrobe, try on clothes virtually, and make better-informed purchasing decisions.

AI, Computer Vision, Virtual Try-On

Hybrid Social Group Optimization for COVID-19 Detection — Publication

Developed a meta-heuristic method for COVID-19 detection from chest X-ray images, achieving over 99% accuracy. Published in the peer-reviewed Cognitive Computation journal.

Machine Learning, Computer Vision

Lung Disease Prediction — Machine Learning

Developed an automated application to detect and predict the onset of lung diseases from X-ray images, achieving over 99% accuracy.

Machine Learning, Medical Imaging

EDUCATION

Bachelor of Technology Maharaja Agrasen Institute of Technology, Delhi — 2016 - 2020

CGPA: 8.3

RECOGNITION

- **Winner, IONathon XTP 1.0** - ION Trading hackathon for breakthrough ideas
- **Top 0.1% Performer** - ION Trading
- **Inspire Program Fellow** - Academic achievement recognition
- **Top 10, Inaugural HackBMU Hackathon** - Illegal Mining Detector
- **99.9 Percentile, JEE Advanced** - Academic achievement